

Spatial-Aware Deep Learning for Reaction Property Prediction

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1. Introduction

Predicting chemical reaction properties has a wide range of applications, including the design of new materials, the discovery of novel drugs, and the optimization of industrial processes. Automating predictions can save costs and drastically reduce the time required in the drug development cycle [1, 2]. Traditionally, this has been accomplished with rule-based expert systems that manually or algorithmically encode reaction heuristics, but these systems are limited because they are not generalizable [1]. The other approach is to use quantum mechanical simulations, like density functional theory (DFT), but the computational cost is prohibitively expensive, especially in systems with large numbers of atoms [1, 3]. In recent years, the creation of large chemical datasets [4] has led to greater interest in using machine learning (ML) to predict chemical properties [5]. ML methods are generalizable because they learn universal lower dimensional embeddings of reactions and are not restricted by the chemical space of rule-based systems [6]. Furthermore, ML methods have significantly lower computational overhead than quantum methods, reducing the computation time for predicting reaction properties from hours to milliseconds. A conventional approach uses template libraries to encode reactions into their molecular fingerprints before feeding them into an ML model [7-10]. However, template-based methods are limited for several inherent reasons: (1) they cannot describe novel reactions, (2) they must choose between generalisation and specificity, and (3) current algorithms neglect the global chemical environment of molecules.

Recent research has turned to template-free ML models that learn reaction representations adaptively. These models can be divided into graph-based and sequence-based models. Graph neural networks (GNNs) treat molecules as graphs and aggregate atom and bond information to obtain a reaction encoding [11-14]. On the other hand, sequence-based models reframe the problem as a machine translation task by using text representations of the reactants and products, usually the simplified molecular-input line-entry system (SMILES) [15, 16]. This enables one to use sophisticated language models based on the powerful transformer architecture [17] to predict chemical reaction properties, which can achieve impressive results [18, 19]. However, sequence-methods are invariably limited because they neglect chemical domain knowledge and are blind to additional atom and bond features. Moreover, SMILES representations are not bijective mappings to molecular structures, meaning they are not permutation invariant.

By treating reactions as graphs or as text, both methods neglect 3D spatial information. Some research has explored incorporating 3D coordinates as atom information in molecular property prediction [11, 20]. However, no previous work has successfully utilised 3D information in reaction property prediction. Furthermore, present research simply uses spatial information as an additional atom feature, and there are few architectures that effectively integrate spatial information into the model itself.

In this work, we propose SpatialRXNN, Spatial Reaction Neural Network, to predict chemical reaction properties. SpatialRXNN first builds a condensed graph of reaction (CGR), which is a superposed graph of reactant and product molecules [21, 22]. A directed-message passing neural network (D-MPNN), which is a type of GNN, aggregates local neighbourhood information on the

reaction graph to generate an encoding for each atom. We then feed these encodings into a transformer. This leverages the representational power of the transformer, but maintains the atom and bond features from the reaction graph and is permutation invariant. Moreover, the inclusion of the transformer unlocks the possibility of embedding spatial information in the positional encodings of each atom, enabling the model to learn spatial relationships between atoms that may not necessarily be captured by the molecular graph. Finally, by analysing the attention weights in the transformer, SpatialRXNN is interpretable and can help generate chemical insights. This is summarised in figure 1. Our main contributions are as follows:

1. We present a novel architecture that synthesises GNNs and transformers to learn both low-level neighbourhood features and high-level global interdependencies for chemical reaction property prediction.
2. We propose a method to integrate spatial information of reactions into the structural representation of the model.
3. We demonstrate that integrating spatial features achieves state-of-the-art results in three reaction property regression datasets.

2. Related Work

Graph Neural Networks. GNNs have become the most ubiquitous ML technique in chemical property prediction because information flow is governed by the underlying graph structure of the molecule [12, 14, 23]. However, GNNs aggregate information in a local neighbourhood of each atom, which limits them from capturing long-range dependencies in the molecule. This is crucial since some atom pairs may be topologically distant but have important interactions, such as in intramolecular hydrogen bonding. One way to address this is to increase the depth of the GNN, but this may lead to over-smoothing and poor performance [24]. Another method is to construct a virtual supernode [25] or draw virtual edges between every pair of atoms [12], but this bypasses the intrinsic structure of the molecule and creates an averaging effect. Instead, more recent work has successfully used the attention mechanism to learn global patterns in molecules [26-28].

Transformers. The transformer [17], originally built for natural language processing (NLP), uses multi-headed self-attention and has seen extensive application in a variety of tasks. Recently, a flood of research has explored using off-the-shelf transformer architectures to predict chemical properties from SMILES strings [18, 19, 29-34]. These models are successful because they apply multiple layers of self-attention that allow every atom to attend to every other atom in the reaction. However, the use of SMILES strings as input neglects chemical domain knowledge and is not permutation invariant. In contrast, Maziarka et al. [35] treat molecules as a list of atoms, and augment the attention mechanism in the transformer to also consider atom features, distances and adjacency. They show that this attains a better performance than using SMILES strings.

3D features in reactions. Reaction properties depend on spatial distances and directions in 3D space, which cannot be captured by SMILES strings or graphs. In molecular property prediction, some research has explored supplementing GNNs with 3D information such as bond distances, bond angles and torsion angles [20, 36, 37]. These studies consistently show that using 3D features improves the accuracy of molecular property prediction. Recently, Ying et al. [38] showed that by

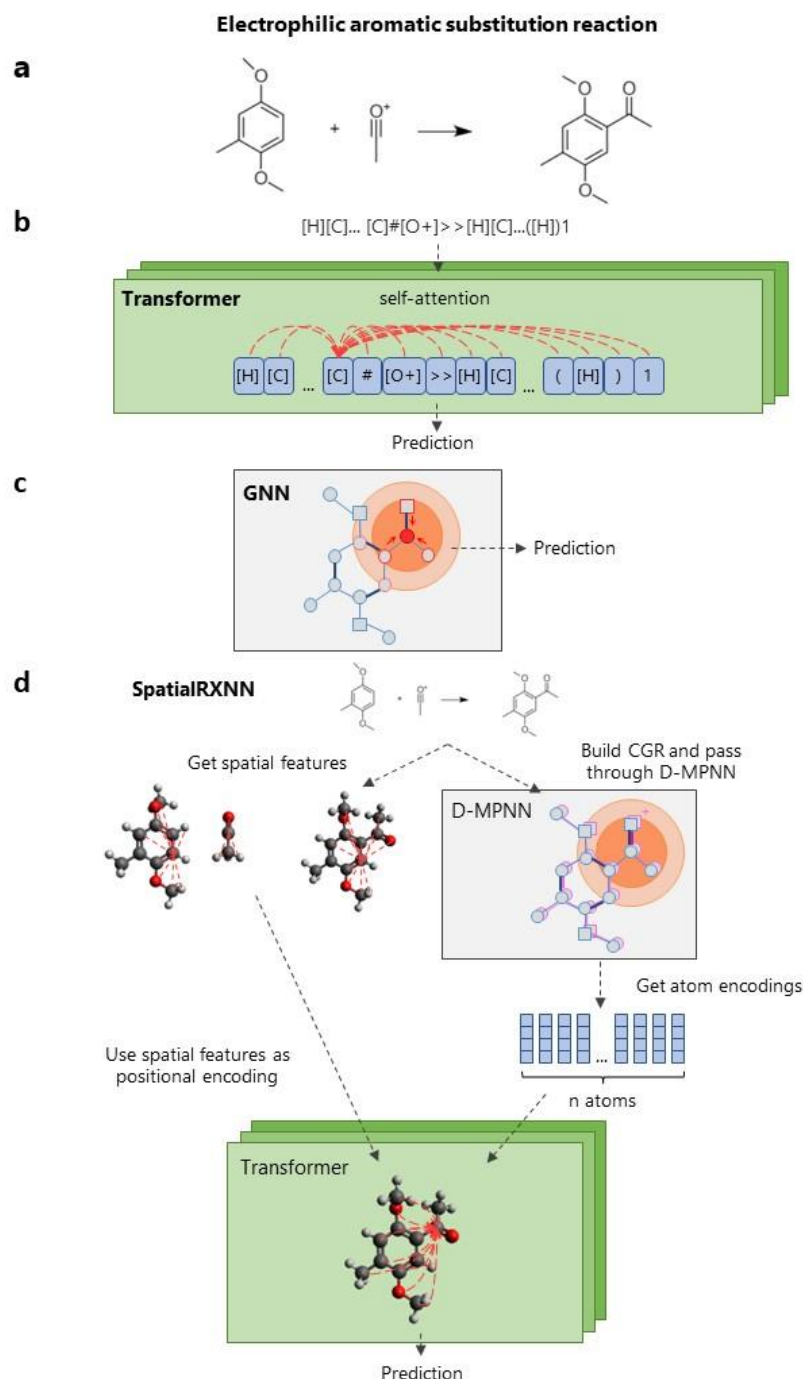


Figure 1. Comparison of SpatialRXNN architecture to previous work **(a)** Acylation of methylhydroquinone dimethyl ether. Steric hindrance by the methoxyl groups may affect reactivity, which is not captured in graphs or SMILES strings. [39] **(b)** Direct application of transformers that takes SMILES strings as input **(c)** GNNs can learn local features on the molecular graph. The concentric circles represent neighbourhood information that is aggregated at higher time-steps. **(d)** Overview of the SpatialRXNN architecture.

replacing the positional encoding of text with a well-designed structural encoding of a graph, transformers could represent graphs. This is crucial because it enables one to also integrate the 3D spatial encoding of a molecule into the transformer [40]. Thus far, no work has been conducted on

using 3D features for reaction property prediction. Unlike molecular property prediction, reactions involve at least two different molecules as it includes reactants and products. Furthermore, there may be multiple reactants and products. Thus, it is not straightforward to use 3D information of reactions. However, 3D features are especially important for reactions since they can predict effects like steric hindrance that may significantly influence reaction properties. Hence, we extend the work by Shi et al. [40] for reaction property prediction.

3. SpatialRXNN

Graph representation. For reaction property prediction, Heid and Green [21] achieved state-of-the-art results by feeding the condensed graph of reaction (CGR) [22] into a D-MPNN. The CGR uses the atom-mapping of a reaction to superpose the graph of reactant atoms onto the graph of product atoms. Following the recommendation of Heid and Green, we construct the atom and bond features of the CGR as the concatenation of the reactant features and the difference between the reactant features and product features. Each atom retains the bonds from both the reactant graph and the product graph. This is illustrated in figure 2. The D-MPNN architecture used is based on the chemprop library developed by Yang et al. [14]. Briefly, each edge aggregates information from neighbouring edges at the previous time step:

$$h_{vw}^{t+1} = \tau(h_{vw}^0 + \mathbf{W} \sum_{k \in \{N(v) \setminus w\}} h_{kv}^t)$$

where h_{vw}^t are the edge features for the edge between nodes v and w at message-passing timestep t , τ is a non-linear activation function, $N(v) \setminus w$ is the set of neighbours of v excluding w , and $\mathbf{W} \in \mathbb{R}^{d \times d}$, d being the size of the hidden dimension. Further details of the D-MPNN can be found in appendix A.

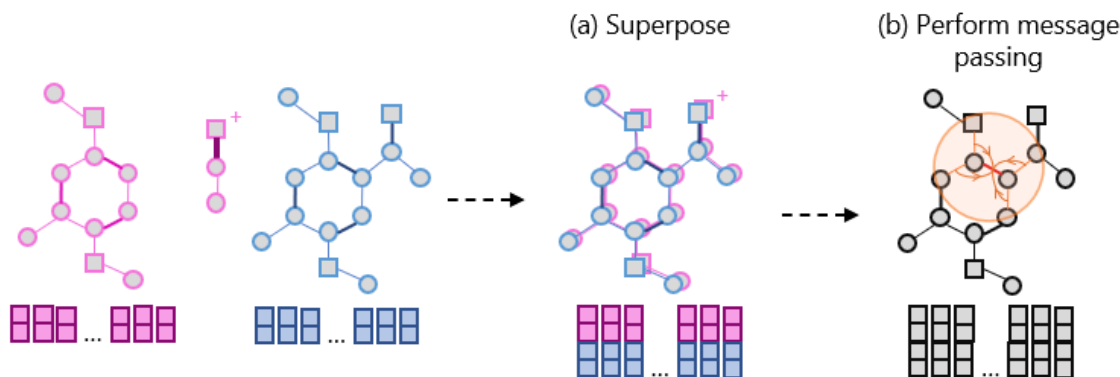


Figure 2. Illustration of CGR and D-MPNN. The reactants are in pink, and the products are in blue. By constructing the CGR, we can treat the reaction as a single graph, which we pass into a D-MPNN to obtain the atom encodings.

Spatial representation. We use RDKit to obtain 3D coordinates of every atom. However, we cannot directly use these coordinates because the spatial features should be invariant to rotation and translation of the molecule. Thus, we construct a matrix of the Euclidean distance between every pair of atoms in the same molecule. For atoms that are in different molecules, we set the distance to be 0.

$$d_{ij} = \|r_i - r_j\|_{L2}$$

Reaction properties depend on changes in molecular structure since structures that are static in the reactants and products do not contribute to the reaction. To model such changes, we evaluate how the distances between equivalent atom pairs change when they go from reactants to products. Thus,

we construct the distance matrix for the CGR as the difference in the distances in the product matrix and the distances in the reactant matrix.

$$\Delta_{diff} = \mathbf{d}_{prod} - \mathbf{d}_{reac}$$

To obtain spatial encodings, we use Gaussian basis kernel functions [41]. Since the relative position of atoms is highly dependent on the size of their atomic radii, we first perform an affine transformation on the distances based on the atom types a_i and a_j of atoms i and j , $\gamma_{(a_i, a_j)}\Delta_{ij} + \beta_{(a_i, a_j)}$ where γ and β are learnable scalars for each pair of atom types. We then pass this through a Gaussian density function to obtain the 3D spatial encoding for each kernel:

$$\psi_{ij}^k = \frac{1}{\sigma^k \sqrt{2\pi}} \exp\left(-\frac{(\gamma_{(a_i, a_j)}\Delta_{ij} + \beta_{(a_i, a_j)} - \mu^k)^2}{2(\sigma^k)^2}\right)$$

where k is the kernel number, σ^k is the learnable kernel centre of the k -th kernel, and μ^k is the learnable scaling factor of the k -th kernel. Further details can be found in appendix A.

Transformer readout. The D-MPNN learns local neighbourhood features in the CGR. We extract the final atom representations after 3 message-passing steps. In NLP, the BERT architecture uses a special [CLS] token to capture the representation of the entire sentence. Similarly, we append a virtual [CLS] node to represent the entire reaction. It is initially set to the mean of all atoms. We integrate the spatial encodings of the reaction into both the absolute positional encoding [17] and relative positional encoding [42] of the transformer. The absolute

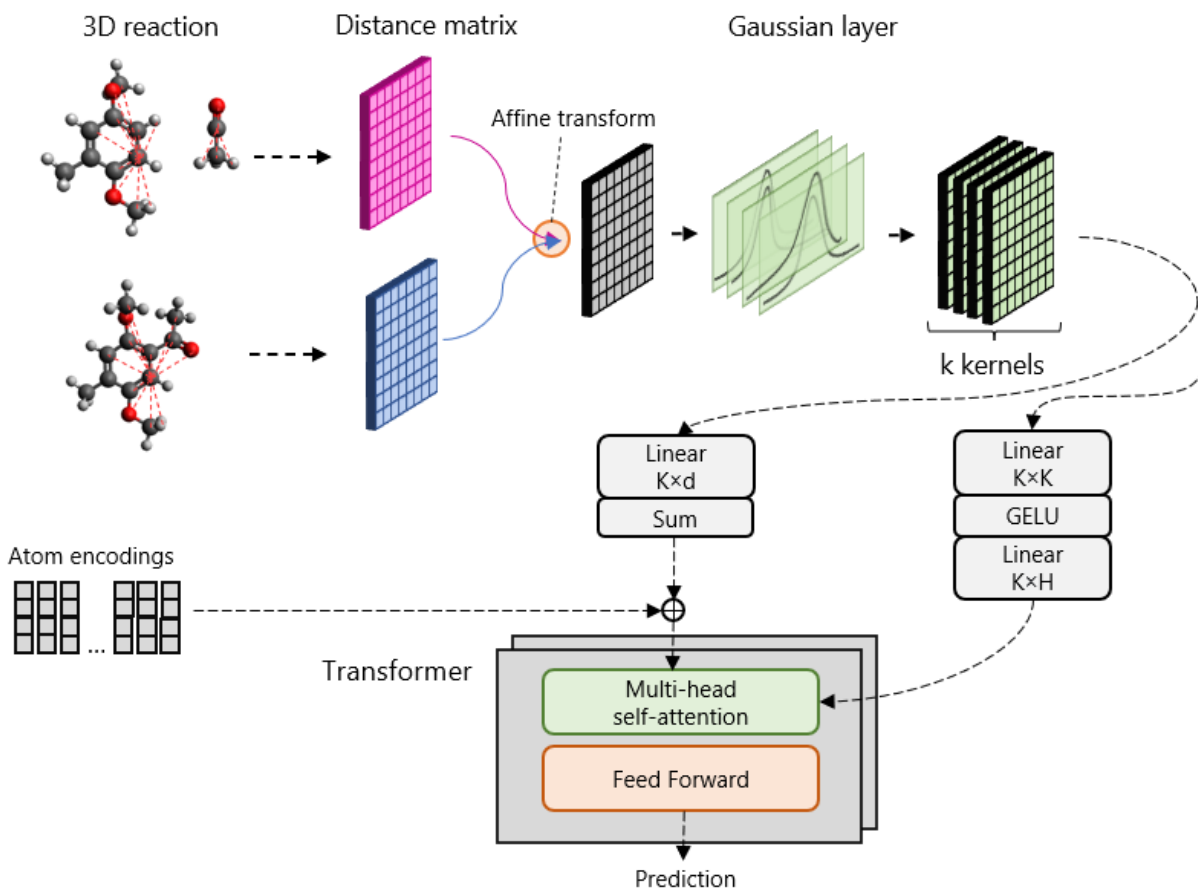


Figure 3. Illustration of how 3D spatial features are obtained from the 3D reaction. Each kernel in the Gaussian layer learns a spatial 'feature'. The atom encodings are obtained from the D-MPNN.

spatial encoding is the sum of 3D spatial encodings to all other atoms, which we add to the input. The relative spatial encoding is a linear projection of the 3D spatial encodings, which is used as an attention bias in the self-attention step. This is summarised in figure 3. Further details can be found in appendix A.

4. Experiments

We conduct experiments on our model with four reaction property regression datasets. These datasets comprise three different tasks: predicting activation energies [3, 43], reaction enthalpies [44], and reaction rate constants [45]. Details on the datasets can be found in appendix C. We compare our model to the baselines and state-of-the-art established by Heid and Green [21]. This is shown in table 1. Each experiment was conducted with five-fold cross validation to obtain the average score and standard deviations. Further implementation and experimental details can be found in appendix B.

	E_a ωB97X-D3 (kcal/mol)	E_a E2/SN2 (kcal/mol)	ΔH Rad-6-RE (eV)	log(k) rate constants (unitless)
Template-based methods				
Morgan Diff default	13.08 ± 0.98	4.75 ± 0.24	1.11 ± 0.03	0.88 ± 0.09
Morgan Diff opt	11.39 ± 0.39	4.53 ± 0.23	0.72 ± 0.02	0.75 ± 0.08
ISIDA default	9.21 ± 0.55	3.14 ± 0.09	0.48 ± 0.02	0.73 ± 0.06
ISIDA opt	7.55 ± 0.48	3.00 ± 0.10	0.43 ± 0.03	0.59 ± 0.06
Sequence methods				
Bert default	14.73 ± 0.52	3.54 ± 0.12	0.76 ± 0.02	0.99 ± 0.03
Bert opt	10.94 ± 0.29	3.37 ± 0.10	0.65 ± 0.01	0.88 ± 0.02
Graph methods				
Grambow default	6.35 ± 0.26	2.76 ± 0.08	0.40 ± 0.01	1.00 ± 0.14
Grambow opt	5.26 ± 0.15	2.86 ± 0.07	– ^a	0.76 ± 0.26
CGR default	4.84 ± 0.29	2.64 ± 0.10	0.16 ± 0.01	0.66 ± 0.29
CGR opt	4.25 ± 0.19	2.65 ± 0.09	0.13 ± 0.01	0.66 ± 0.24
SpatialRXNN (ours)	2.68 ± 0.61	2.53 ± 0.11	0.14 ± 0.02	0.33 ± 0.08

Table 1: Comparison of mean absolute error (MAE) score on each dataset. Lower score is better. The best performance per dataset is highlighted in bold. The root mean squared error (RMSE) scores and model sizes can be found in appendix D. ^aAs highlighted by Heid and Green, grambow opt simply memorises enthalpies of molecules that repeat in the training and test set, thus we exclude it from this comparison.

Discussion. SpatialRXNN outperforms previous methods in predicting activation energies and reaction rates. However, SpatialRXNN does not observe significant improvement in predicting reaction enthalpies. This could be because reaction enthalpies are mostly dependent on bond enthalpies and not on the 3D environment, making global interactions and spatial features less significant. Thus, a simple GNN which can learn bond enthalpies would be able to predict reaction enthalpies. We conduct further ablation studies to evaluate the contributions of each component of our model. This can be found in appendix D.

The transformer component of SpatialRXNN also allows us to obtain attention weights, revealing which atom neighbourhoods influence the final prediction the most. Some examples are given in figure 4. This makes SpatialRXNN more interpretable than GNNs, which increases trust in the model prediction by allowing chemists to verify if the prediction aligns with chemical principles. Furthermore, it may facilitate the discovery of new chemical and physical insights by unveiling unexpected mechanisms. Finally, it speeds up optimization of chemical processes by identifying key substructures in the molecules which contribute to the property of interest.

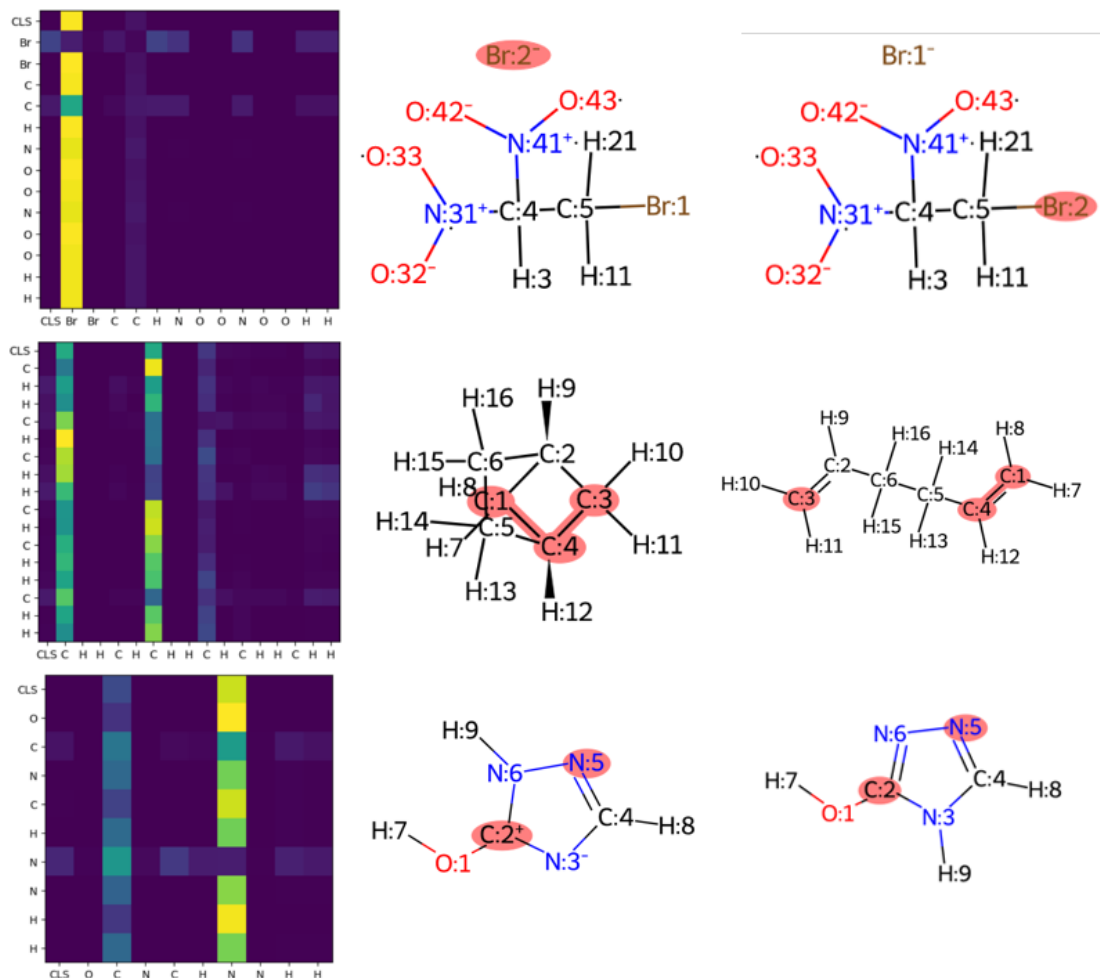


Figure 4. Attention heat-maps of the last layer of the transformer predicting computational activation energy. The first reaction is from e2sn2, while the bottom two are from ω B97X-D3. The molecule on the left is the reactant while the molecule on the right is the product. The label next to each atom is its mapping between the reactant and product. We highlight atoms that the [CLS] token attends to the most.

5. Conclusions

In this work, we present the SpatialRXNN as an effective architecture for reaction property prediction. SpatialRXNN can represent local and global interactions and incorporates 3D spatial information into the structure of the model, achieving state-of-the-art results in predicting activation energies and reaction rate constants. Moreover, SpatialRXNN is interpretable, which (1) allows its predictions to be verified, (2) may reveal chemical insights, and (3) speeds up optimization of chemical processes.

However, SpatialRXNN also has limitations. Compared to previous architectures, SpatialRXNN requires more data pre-processing steps to generate atom-mapping and 3D information. Furthermore, if the atom-mapping or 3D information is noisy or incomplete, it may hinder model performance. Nonetheless, our work highlights that spatial features are critical for reaction property prediction, especially for kinetic properties, and that future work should explore building models that can make greater use of spatial information like bond angles.

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Appendix A – Architecture details

Building the CGR. Let the molecular graph be an undirected graph G with atoms V and bonds E . Each atom $v \in V$ has a feature vector $x_v \in \mathbb{R}^a$ and each bond $(u, v) \in E$ from atom u to v has a feature vector $e_{vw} \in \mathbb{R}^b$. To form the CGR, we superpose the atoms from the products onto the corresponding atoms from the reactants based on the atom-mapping. We concatenate the features from the reactant graph with the difference in features in the reactant and product graphs to obtain the new features:

$$\begin{aligned}x_v &= \text{cat}(x_v^{\text{reac}}, x_v^{\text{diff}}) \\ e_{vw} &= \text{cat}(e_{vw}^{\text{reac}}, e_{vw}^{\text{diff}})\end{aligned}$$

The list of bonds now comprises bonds from both the reactant graph G^{reac} and product graph G^{prod} , $E = E^{\text{reac}} \cup E^{\text{prod}}$. All four datasets used are balanced, so the list of atoms is simply $V = V^{\text{reac}} = V^{\text{prod}}$.

D-MPNN. The D-MPNN consists of a message passing phase and a readout phase. The message passing phase consists of T time steps, where each edge is updated based on the information from neighbouring edges. We initialize the hidden state in each edge at time step 0 by concatenating the features of the first atom x_v to the bond features e_{vw} and passing it through a single layer:

$$h_{vw}^0 = \tau(\mathbf{W}_i \text{cat}(x_v, e_{vw}))$$

where $\mathbf{W}_i \in \mathbb{R}^{d \times d_i}$, and d being the hidden size (300), d_i being the size of $\text{cat}(x_v, e_{vw})$, and $\tau()$ being a nonlinear activation function (ReLU). The message m_{vw}^{t+1} at time step $t + 1$ in edge vw is the sum of hidden states from neighbouring edges directed towards v at the previous time step t :

$$m_{vw}^{t+1} = \sum_{k \in \{N(v) \setminus w\}} h_{kv}^t$$

where $N(v) \setminus w$ is the set of neighbours of v excluding w . To obtain the new hidden state values:

$$h_{vw}^{t+1} = \tau(h_{vw}^0 + \mathbf{W}_m m_{vw}^{t+1})$$

where $\mathbf{W}_m \in \mathbb{R}^{d \times d}$. We repeat this for 3 times steps. Finally, in the readout phase, we transform the hidden states back into atom representations:

$$h_v = \tau(\mathbf{W}_0 (\text{cat}(x_v, \sum_{w \in N(v)} h_{wv}^T)))$$

where $\mathbf{W}_0 \in \mathbb{R}^{d \times d_0}$, where d_0 is the size of $\text{cat}(x_v, h_{wv})$. At the end, we obtain representations h_v of each atom in the CGR of shape $n \times d$, where n is the number of atoms.

Getting the spatial features. We use RDKit to generate the 3D coordinates of each atom in each molecule in the reaction. Within each molecule, we calculate the Euclidean distance between every pair of atoms d_{ij} . For atoms in different molecules, we set the distance to 0. We repeat this for the reactants and the products separately.

$$d_{ij} = \begin{cases} \|r_i - r_j\|_{L2}, & \text{if same molecule} \\ 0, & \text{otherwise} \end{cases}$$

We first pad the distance matrices to the maximum number of atoms and append the virtual [CLS] atom. To find the distance matrix for the CGR, we take the difference in the distances in the product matrix and the distances in the reactant matrix.

$$\Delta_{\text{diff}} = \mathbf{d}_{\text{prod}} - \mathbf{d}_{\text{reac}}$$

We perform an affine transformation on the distances based on the atom types a_i and a_j of atoms i and j , $\gamma_{(a_i,a_j)}\Delta_{ij} + \beta_{(a_i,a_j)}$ where γ and β are learnable scalars for each pair of atom types. We then pass this distance matrix through a set of gaussian density functions with k kernels:

$$\psi_{ij}^k = \frac{1}{\sigma^k \sqrt{2\pi}} \exp\left(-\frac{(\gamma_{(a_i,a_j)}\Delta_{ij} + \beta_{(a_i,a_j)} - \mu^k)^2}{2(\sigma^k)^2}\right)$$

where k is the kernel number, σ^k is the learnable kernel centre of the k -th kernel, and μ^k is the learnable scaling factor of the k -th kernel. For the virtual [CLS] atom, and all other padded atoms, we set ψ_{ij}^k to be 0. The shape of ψ is $N \times N \times K$. The absolute spatial encoding of atom i is calculated as the sum of the spatial encodings to all other atoms j and passed through a linear layer:

$$\varphi_i^{sum} = \mathbf{W}_k^0 \sum_{j \in V} \psi_{ij}^k$$

where $\mathbf{W}_k^0 \in \mathbb{R}^{d \times K}$. The shape of φ^{sum} is $N \times d$, where N is the maximum number of atoms (after padding) and d is the hidden size. The relative spatial encoding between atoms i and j is a linear projection of the 3D spatial encodings:

$$\phi_{ij}^{spatial\ encoding} = GELU(\psi_{ij} \mathbf{W}_k^1) \mathbf{W}_k^2$$

where $\mathbf{W}_k^1 \in \mathbb{R}^{K \times K}$ and $\mathbf{W}_k^2 \in \mathbb{R}^{K \times H}$, and H is the number of heads in the transformer. The final shape of ϕ is $N \times N \times H$.

Transformer readout. We take the final atom representations from the D-MPNN $\mathbf{h}$ and pad it to the maximum number of atoms N . We also append a virtual [CLS] atom to represent the entire reaction. We first briefly introduce the transformer architecture. The transformer consists of N attention layers, where each layer is composed of a multi-head self-attention block, followed by a feed-forward block, with both blocks having residual connections and layer normalization. The multi-headed self-attention comprises H heads. Let the input to the l -th layer be $\mathbf{X}^l$. Head h computes the vectors $\mathbf{Q}_i^{l,h} = \mathbf{X}^l \mathbf{W}_Q^{l,h}$, $\mathbf{K}_i^{l,h} = \mathbf{X}^l \mathbf{W}_K^{l,h}$, $\mathbf{V}_i^{l,h} = \mathbf{X}^l \mathbf{W}_V^{l,h}$. The attention operation is:

$$Attention(\mathbf{Q}_i^{l,h}, \mathbf{K}_j^{l,h}, \mathbf{V}_j^{l,h}) = softmax\left(\frac{\mathbf{Q}_i^{l,h} (\mathbf{K}_j^{l,h})^T}{\sqrt{d_k}}\right) \mathbf{V}_j^{l,h}$$

Our implementation of the transformer encoder is inspired by The Annotated Transformer¹.

We augment the transformer by adding absolute positional encoding and relative positional encoding. The absolute spatial encoding is added to the representation for each atom after appending the CLS token:

$$\begin{aligned} h_{CLS} &= mean(\mathbf{h}) \\ \mathbf{X} &= pad(cat(h_{CLS}, \mathbf{h})) \\ X'_i &= X_i + \varphi_i^{sum} \end{aligned}$$

The relative spatial encoding is added as an attention bias in the self-attention step:

$$Attention(\mathbf{Q}_i^{l,h}, \mathbf{K}_j^{l,h}, \mathbf{V}_j^{l,h}) = softmax\left(\frac{\mathbf{Q}_i^{l,h} (\mathbf{K}_j^{l,h})^T}{\sqrt{d_k}} + \phi_{ij}^{h,spatial\ encoding}\right) \mathbf{V}_j^{l,h}$$

Finally, we extract the embeddings of the [CLS] token and pass it through a feedforward network to obtain the final prediction.

¹ <http://nlp.seas.harvard.edu/annotated-transformer/>

Appendix B – Experimental details

Training details. All models are trained on a single GPU for 100 epochs. We first pretrain the D-MPNN by letting the [CLS] representation bypass the transformer. Then, we add the transformer as an adapter module. We find that this stabilizes training and allows the model to reach a lower loss. The batch size is 50. The model dropout is set to 0. The size of the hidden dimension in the D-MPNN and transformer is 300. We use two layers in the final feed forward block. The learning rate increased linearly from 10^{-4} to 10^{-3} for two warmup epochs, then decreased exponentially from 10^{-3} to 10^{-4} . We use the Adam optimizer. The transformer has 2 layers and each layer has 3 self-attention heads. We use 60 gaussian kernels.

Atom and bond features used are the same as in [14].

Atom Features Used.

Feature	Description	Size
Atomic number	Type of atom by atomic number	100
Degree	Number of bonds attached to atom	6
Formal charge	Electronic charge assigned to atom	5
chirality	Unspecified, tetrahedral CW/CCW, other	4
# Hs	Number of bonded hydrogen atoms	5
Hybridization	Sp, sp ² , sp ³ , sp ³ d, sp ³ d ²	5
Aromaticity	Whether this atom is part of aromatic system	1
Atomic mass	Mass of atom divided by 100	1

Bond Features Used.

Feature	Description	Size
Bond type	Single, double, triple, aromatic	4
conjugated	Whether the bond is conjugated	1
In ring	Whether the bond is part of a ring	1
stereo	None, any, E/Z, cis/trans	6

Obtaining 3D Information. We obtain 3D data with RDKit². Conformers are isomers of a molecule that differ from another isomer by rotation of a single bond in the molecule. Each molecule may have multiple conformers in 3D space. We generated 3D conformers with the Experimental-Torsion Knowledge Distance Geometry method (ETKGD) [46] and optimize with the Merck Molecular Force Field (MMFF94) [47] in RDKit. For each molecule, we randomly generate 200 conformations and select the one with the lowest energy. This is because the DFT calculations in the ω B97X-D3 dataset also use the lowest energy conformer. For molecules that cannot optimize with MM6FF94, we then try optimizing with the Universal Force Field (UFF).

However, we found that RDKit was unable to generate conformers for some very strained molecules. We provide some examples in figure 5. For reactions with such molecules, we simply set all spatial distances to 0. This negates the effect of the spatial encoding in the model. Future work should look at alternative methods to handle such molecules. Nonetheless, these molecules are so strained that they make up a small proportion of the datasets (< 1% of reactions) and are unlikely to appear in real-world scenarios, thus it is also possible to remove these molecules from

² <https://www.rdkit.org/docs/index.html>

the dataset entirely. However, we include them to more fairly compare our results to previous work.

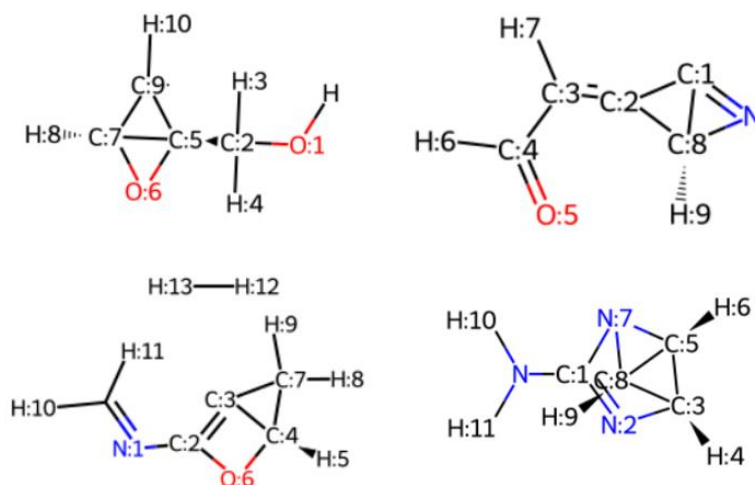


Figure 5. Examples of molecules that RDKit was unable to generate 3D conformers for.

Appendix C – Dataset details

All datasets were obtained from [48]. All datasets are balanced and contain explicit hydrogens.

ω B97X-D3. ω B97X-D3 contains computational activation energies of forward reactions at the ω B97X-D3/def2-TZVP level of theory as generated by [3]. It contains 11961 reactions. It includes reactions transforming unimolecular reactants into unimolecular or multimolecular products. Before training and testing on ω B97X-D3, we first pre-train the model on 16365 reactions at the B97-D3/def2-mSVP level of theory.

E2/SN2. E2/SN2 contains computational activation energies for competing E2 and SN2 reactions at the MP2/6-311G(d) level of theory as generated by [43]. It contains 3626 reactions.

Rad-6-RE. The Rad-6-RE database contains computational reaction enthalpies as generated by [44]. It contains 63,849 reactions.

Log(k). The log(k) rate constant dataset contains reaction rate constants as generated by [45]. The labels were set to the logarithm of the rate constants at 1000K, $\log\left(\frac{k(1000K)}{k_{ref}}\right)$, where the units for k are $\text{cm}^3\text{mol}^{-1}\text{s}^{-1}$ for bimolecular reactions and s^{-1} for unimolecular reactions. $k_{ref} = 1$ in the same units.

Appendix D – Additional Results

	<i>Ea</i> ωB97X-D3 (kcal/mol)	<i>Ea</i> E2/SN2 (kcal/mol)	ΔH Rad-6-RE (eV)	log(k) rate constants (unitless)
Template-based methods				
Morgan Diff default	18.27 $\pm$ 0.98	6.67 $\pm$ 0.35	1.44 $\pm$ 0.04	1.44 $\pm$ 0.09
Morgan Diff opt	16.10 $\pm$ 0.40	6.24 $\pm$ 0.21	1.02 $\pm$ 0.06	1.20 $\pm$ 0.16
ISIDA default	13.63 $\pm$ 0.66	4.27 $\pm$ 0.07	0.92 $\pm$ 0.04	1.26 $\pm$ 0.23
ISIDA opt	11.80 $\pm$ 0.78	4.10 $\pm$ 0.17	0.76 $\pm$ 0.02	1.03 $\pm$ 0.08
Sequence methods				
Bert default	19.12 $\pm$ 0.66	4.75 $\pm$ 0.12	0.99 $\pm$ 0.03	1.40 $\pm$ 0.27
Bert opt	15.34 $\pm$ 0.36	4.56 $\pm$ 0.11	0.88 $\pm$ 0.02	1.24 $\pm$ 0.29
Graph methods				
Grambow default	9.11 $\pm$ 0.51	3.74 $\pm$ 0.13	0.55 $\pm$ 0.03	1.00 $\pm$ 0.14
Grambow opt	7.98 $\pm$ 0.37	3.83 $\pm$ 0.11	–	0.76 $\pm$ 0.26
CGR default	7.63 $\pm$ 0.43	3.59 $\pm$ 0.09	0.28 $\pm$ 0.02	0.66 $\pm$ 0.29
CGR opt	6.88 $\pm$ 0.38	3.61 $\pm$ 0.07	0.25 $\pm$ 0.02	0.66 $\pm$ 0.24
SpatialRXNN (ours)	4.39 $\pm$ 1.00	3.40 $\pm$ 0.15	0.23 $\pm$ 0.03	0.55 $\pm$ 0.32

Table 2: Comparison of root mean squared error (RMSE) score on each dataset. Lower score is better. The best performance per dataset is highlighted in bold.

	<i>Ea</i> ωB97X-D3	<i>Ea</i> E2/SN2	ΔH Rad-6-RE	log(k) rate constants
Template-based methods				
Morgan Diff default	307,801	307,801	307,801	307,801
Morgan Diff opt	8,224,801	7,652,101	8,224,801	8,224,801
ISIDA default	13,159,801	55,801	4,291,501	200,101
ISIDA opt	72,747,201	334,801	8,943,601	7,363,201
Sequence methods				
Bert default	77,401	77,401	77,401	77,401
Bert opt	6,381,601	361,201	6,381,601	6,381,601
Graph methods				
Grambow default	361,801	361,801	361,801	361,801
Grambow opt	24,877,601	16,754,401	20,035,401	8,269,801
CGR default	378,601	378,601	378,601	378,601
CGR opt	2,817,101	2,817,101	10,371,601	6,393,701
SpatialRXNN (ours)	1,489,516	1,489,516	1,489,516	1,489,516

Table 3: Number of parameters for each model per dataset.

Ablation studies.

	MAE	RMSE
Ignore atom-mapping information	11.17	14.53
Virtual edges between mapped atoms	7.60	10.43
Build CGR with atom-mapping	4.15	6.38

Table 4: Ablation study on use of atom-mapping. We train the CGR and D-MPNN model alone on the ωB97X-D3 dataset with pretraining on B97-D3.

	MAE	RMSE
CGR w/o spatial features or transformer	4.15	6.38
Spatial features w/o transformer	4.00	6.28
Spatial features w/ transformer	3.87	6.22

Table 5: Ablation study on contributions of adding spatial features and the transformer.

We conduct ablation studies on the ωB97X-D3 dataset to evaluate the contributions of each component of our model. The models are trained for 100 epochs. In table 2, we compare not using atom-mapped information, drawing virtual edges between mapped-atoms, or building a CGR. We find that the CGR has a stronger inductive bias than drawing virtual edges, and that using atom-

mapped information is crucial in predicting reaction properties. In table 3, we compare the effects of adding spatial features and the transformer. We find that both improve model performance.

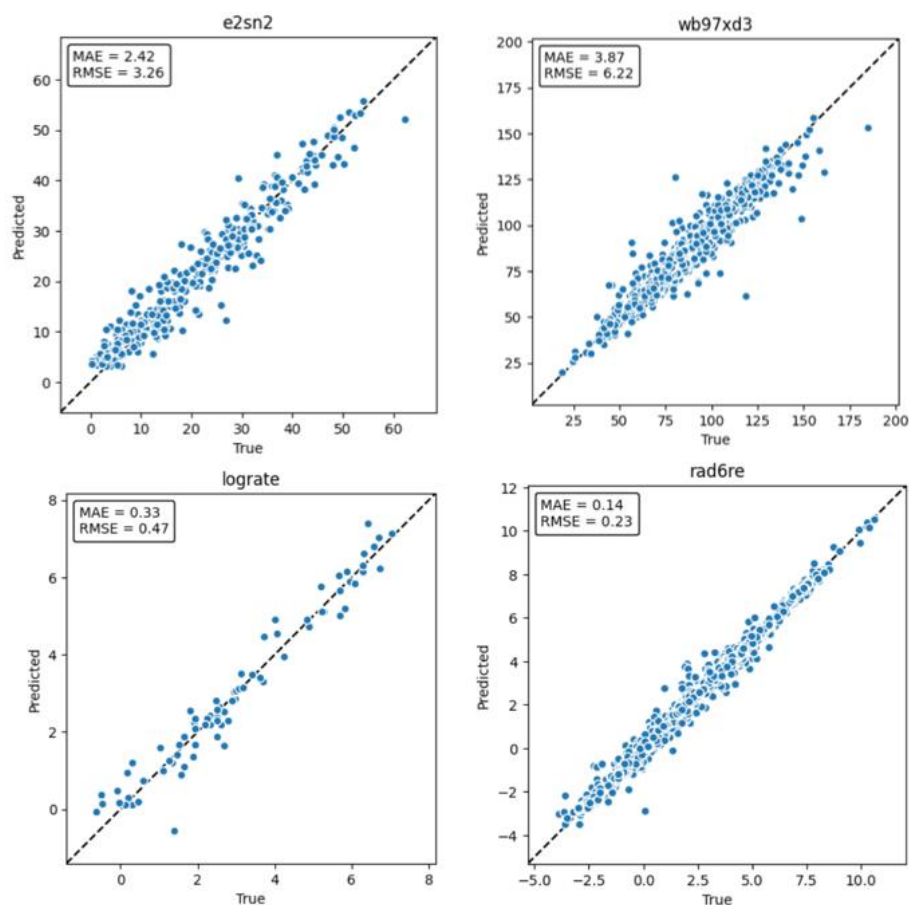


Figure 6: Predictions vs. true values by SpatialRXNN in one fold for each dataset